

India Agricultural Intelligence Dashboard

Crop Production, Yield & State-wise Analytics

1997-2019 | Python + MySQL (2 Tables) + Power BI Desktop

19,480 Records (Excl. Coconut) · 30 States · 54 Crops · 4.41 Avg Yield

TOOL	SOURCE	RECORDS	DATA TYPE
Python + MySQL + Power BI	Govt. of India / Kaggle	19,480 + 245,529 (2 tables)	Time Series (1997-2019)

01 EXECUTIVE SUMMARY

- Processed two complementary Indian agricultural datasets — a 19,689-row state-level file with unique input-factor columns (rainfall, fertilizer, pesticide, yield) and a 246,091-row district-level file with 4× the crop coverage — building a genuine 2-table MySQL model, the first JOIN-capable architecture in this portfolio.
- Discovered and corrected a severe unit-of-measurement distortion: Coconut is counted in individual nuts rather than weight, accounting for 95.41% of raw "total production" across just 0.87% of records — every production ranking in this dashboard explicitly excludes it.
- Engineered boundary-safe derived features and 15+ DAX measures across a 3-page, 21-visual Power BI dashboard, revealing that fertilizer use rose 53.2% between 1997 and 2019 while average yield rose only 9.9% — a genuine diminishing-returns finding — and that Uttar Pradesh leads national production at 4.44 billion units (excl. coconut).
- Demonstrated that cross-crop correlation analysis is structurally misleading in this dataset (Rainfall/Fertilizer/Pesticide repeat identically across all crops within a state-year) and that isolating a single crop reveals genuine signal — Wheat shows a +0.447 fertilizer correlation and a -0.410 rainfall correlation, both consistent with known agronomic behavior.

02 PROBLEM STATEMENT

Two public Indian agricultural datasets — one offering district-level granularity across 646 districts and 124 crops, the other offering state-level environmental input factors (rainfall, fertilizer, pesticide usage) unavailable anywhere else — existed as separate flat files with no unified reporting layer and no resolution of a critical measurement inconsistency: Coconut production is recorded in individual nut counts while every other crop is recorded in tonnes, a distinction invisible in raw aggregate figures. Agricultural planners had no way to compare states or crops on production volume without this distortion silently dominating every ranking, no mechanism to isolate genuine input-yield relationships from a dataset where fertilizer and rainfall figures are repeated at the state-year level rather than measured per crop, and no consolidated geographic view spanning both district-level detail and state-level environmental context.

03 OBJECTIVES

- Obj 1: Execute a structured Python EDA pipeline across both files — clean, deduplicate, remove incomplete trailing years, and engineer boundary-safe derived features.

- Obj 2: Build a 2-table normalized MySQL database — a primary state-level table with input factors and a secondary district-level table for geographic depth.
- Obj 3: Diagnose and correct the Coconut unit-of-measurement distortion, ensuring every production-based ranking excludes it while preserving it for transparent, dedicated comparison.
- Obj 4: Create a multi-dimension analytical VIEW and DAX measures enabling correctly weighted state, crop, season, and rainfall-bucket comparisons.
- Obj 5: Build a 3-page, 21-visual interactive dashboard spanning national overview, input-yield driver analysis, and district-level geographic deep-dive.
- Obj 6: Surface actionable agricultural intelligence — fertilizer efficiency trends, seasonal crop patterns, and genuine single-crop input correlations — delivered with a portfolio-ready custom JSON theme.

04 DATASET OVERVIEW

Source & Structure

Attribute	File 1 (Primary)	File 2 (Secondary)
Dataset Name	crop_yield.csv	crop_production.csv
Granularity	State-level	District-level
Raw Records	19,689	246,091
Clean Records Used	19,652 (excl. partial 2020)	245,529 (excl. partial 2015)
States / Districts	30 states	33 states · 646 districts
Crops Covered	55	124
Year Range	1997–2019	1997–2014 (usable)
Unique Value	Rainfall, Fertilizer, Pesticide, Yield	Broader crop & district coverage
Null Values	Zero	3,729 blank Production (Area still present)

Critical Data Quality Finding — Coconut Unit Mismatch

Coconut is recorded in individual nut counts rather than weight, causing it to represent 95.41% of File 1's total raw production value (92.07% in File 2) despite comprising under 1% of records. Every production-based ranking in this dashboard uses an explicit "excluding coconut" filter; Coconut is preserved in the underlying tables and shown separately for transparency.

05 TOOLS & TECHNOLOGIES USED

Tool / Technology	Platform	Purpose in This Project
Python 3.11	pandas, numpy, matplotlib, seaborn	Dual-file EDA, boundary-safe feature engineering, 10-chart visualization
MySQL 8.0	MySQL Workbench	2-table schema, dual LOAD DATA imports, 10 analytical queries, 1 VIEW

Tool / Technology	Platform	Purpose in This Project
Power BI Desktop	Microsoft (Latest 2024)	Dual-table connection, DAX modelling, 21-visual dashboard with India map
DAX	Power BI Native Language	Core KPI measures plus weighted-average VIEW correction measure
Govt. of India / Kaggle	Public Dataset Platforms	Two independently-sourced, complementary agricultural datasets
JSON Theme File	Power BI Custom Theme	India Agricultural Intelligence — Harvest Gold portfolio theme

06 DATA PREPARATION — PYTHON & MYSQL PIPELINE

Both files required removal of an incomplete trailing year (a data-collection cutoff artifact, not genuine agricultural data) and whitespace cleanup on categorical fields. A 2-table MySQL model was built to preserve each file's unique strength rather than forcing an artificial merge across mismatched granularities.

Phase 1 — Python (10 Steps)

#	Python Action	Reason / Impact
01	Loaded both CSVs, compared structure	Confirmed genuinely complementary datasets, not duplicate formats
02	Stripped whitespace from Season/Crop/State fields in both files	Both files had trailing spaces (e.g. "Kharif ")
03	Removed 2015 from File 2 (562 rows) and 2020 from File 1 (37 rows)	Both are partial-year data-collection cutoffs, ~1/25th normal row count
04	Verified Coconut unit mismatch quantitatively	Confirmed 95.41% / 92.07% distortion across both files independently
05	Engineered YieldBucket, RainfallBucket, FertilizerPerHectare, YearEra	4 boundary-safe derived features on File 1
06	Generated Charts 1-2 (Overview, Coconut Distortion side-by-side)	Visually proved the unit-mismatch finding
07	Generated Charts 3-5 (Top States, Top Crops, Season Yield)	Uttar Pradesh leads; Rice/Wheat dominate area; Whole Year crops lead yield
08	Generated Charts 6-8 (Fertilizer Trend, Wheat Correlation, Kharif/Rabi)	Diminishing returns pattern; genuine single-crop signal; seasonal validation
09	Generated Charts 9-10 (Top Districts, Correlation Comparison)	District-level geography from File 2; mixed-vs-single-crop correlation contrast
10	Exported both cleaned CSVs for MySQL	19,652 and 245,529 rows respectively, ready for import

Phase 2 — MySQL (2-Table Model, 15 Steps)

#	MySQL Action	Reason / Impact
01	Created crop_yield (primary) and crop_production_district (secondary) tables	2-table model preserving each file's distinct strength
02	Imported crop_yield — 19,652 rows, 0 skipped	13,278 rounding warnings (DECIMAL precision) — expected, not an error
03	First crop_production_district import attempt failed — Error 1366: Incorrect decimal value "	Strict SQL mode rejected a blank-to-decimal conversion at row 47
04	Retried with SET SESSION sql_mode="" and TRIM() — import succeeded, but null count came back 0 instead of expected 3,730	Silent data corruption — investigated further (see Challenge 2)
05	Diagnosed hidden trailing \r character from Windows line endings on the last-loaded field	Root cause confirmed byte-for-byte against the actual source file
06	Rebuilt import via staging column + explicit \r-aware UPDATE	Verified: 245,529 total rows, exactly 3,729 genuine nulls, spot-checked row confirmed NULL
07	Created 8 indexes across both tables	Query performance optimization
08–14	Executed 10 analytical queries across both tables	All results cross-validated against Python's independently generated charts
15	Created vw_crop_yield_summary VIEW — 6-dimension aggregation	Powers Power BI's Input-Yield Drivers page

07 DAX MEASURES CREATED

Measure Name	DAX Formula	Result / Use
Total Records	COUNTROWS(crop_yield)	19,652 — includes coconut intentionally
Total Production (Excl Coconut)	CALCULATE(SUM(production), crop<>"Coconut")	14,845,100,222
Avg Yield (Excl Coconut)	CALCULATE(AVERAGE(yield_value), crop<>"Coconut")	4.41
Avg Fertilizer Per Hectare	AVERAGEX(crop_yield, DIVIDE(fertilizer,area))	Powers the diminishing-returns trend line
States Covered / Crops Covered	DISTINCTCOUNT(state) / DISTINCTCOUNT(crop)	30 / 54
VIEW Weighted Avg Yield	DIVIDE(SUMX(view,avg×count),SUM(count))	Corrects aggregation across the 6-dimension VIEW

08 DASHBOARD VISUALS BUILT (21 Visuals)

Page 1 — National Overview (9 Visuals)

#	Visual Type	Fields Used	Insight Delivered
1–4	KPI Cards	Total Records, Total Production, Avg Yield, States Covered	19,652 records · 14.85bn production · 4.41 avg yield · 30 states
5	Bar Chart	state / Total Production (Excl Coconut)	Uttar Pradesh leads at 4.44 billion
6	Bar Chart	crop / Total Area	Rice and Wheat dominate India's cropped land
7	Filled Map	state / Total Production (Excl Coconut)	Geographic production intensity across India
8–9	Slicers	crop_year (range), season (dropdown)	Filters all Page 1 visuals

Page 2 — Input-Yield Drivers (7 Visuals)

#	Visual Type	Fields Used	Insight Delivered
10	Bar Chart	rainfall_bucket (VIEW) / VIEW Weighted Avg Yield	Yield varies meaningfully by rainfall band
11	Dual-Axis Line Chart	crop_year / Avg Fertilizer Per Hectare + VIEW Weighted Avg Yield	Fertilizer +53% vs Yield +10% — diminishing returns
12	Bar Chart	season / Avg Yield (Excl Coconut)	Whole Year crops (13.23) far exceed single-season crops
13	Donut Chart	yield_bucket (VIEW) / record_count	48.15% of records fall in Medium yield band
14–15	Slicers	crop (dropdown), year_era (VIEW)	Filters Page 2 visuals
16	2 Bar Charts (Kharif/Rabi)	crop / Total Area, filtered by season, Top 5	Kharif: Rice, Cotton, Soyabean, Maize, Groundnut · Rabi: Wheat, Gram, Jowar, Rapeseed&Mustard, Niger seed

Page 3 — Geographic Deep-Dive (5 Visuals)

#	Visual Type	Fields Used	Insight Delivered
17	Table (Top N 10)	state_name, district_name / Sum of production	Muzaffarnagar (UP) leads at 269M
18	Filled Map	state_name / Sum of production	District-table-sourced geographic distribution
19	Table	state_name / Count of crop, Sum of production	Crop diversity by state — 124 crops total coverage
20–21	Slicers	state_name, crop_year (range)	Filters Page 3 visuals

09 KEY INSIGHTS & RESULTS

A. The Coconut Unit-Mismatch Discovery

- Coconut alone accounts for 95.41% of File 1's raw total production value across just 0.87% of records — a distortion confirmed independently in File 2 as well (92.07%), proving it reflects the crop's counting convention (individual nuts) rather than a data error unique to one source.
- Once excluded, a meaningful ranking emerges: Sugarcane genuinely leads production at 7.2 billion, followed by Rice (2.2bn) and Wheat (1.8bn) — figures that would otherwise be invisible behind Coconut's 308 billion nut-count.

B. Production & Area Leadership

- Uttar Pradesh leads all states in production (excl. coconut) at 4.44 billion, nearly 2.4× second-place Maharashtra (1.88bn), reflecting its position as India's most populous agricultural state.
- By area planted — a unit-agnostic metric unaffected by the coconut issue — Rice (992M hectares) and Wheat (602M hectares) dominate India's cropped land, together exceeding the combined area of the next 8 largest crops.

C. Seasonal Patterns Validate Data Quality

- Kharif (monsoon) season's top 5 crops by area — Rice, Cotton, Soyabean, Maize, Groundnut — and Rabi (winter) season's top 5 — Wheat, Gram, Jowar, Rapeseed&Mustard, Niger seed — match real Indian farming calendars precisely, a strong sanity check that the underlying data is trustworthy.
- Whole Year crops average a yield of 13.23 — more than double Winter's 5.29 and five times Kharif's 2.48 — consistent with perennial high-yield crops like sugarcane and banana being classified under this season label.

D. Diminishing Returns on Fertilizer

- Average fertilizer use per hectare rose from 102.94 (1997-2005) to 157.70 (2011-2020) — a 53.2% increase — while average yield rose from 4.19 to 4.60, just 9.9% over the same period.
- This gap suggests fertilizer application has moved well past its most efficient range for much of India's cropped area, an actionable finding for agricultural input-efficiency policy.

E. Why Single-Crop Correlation Matters

- Mixing all 54 crops together produces near-zero correlation between Fertilizer/Rainfall and Yield (0.003 to 0.02) — because these input columns repeat identically across every crop within a state-year rather than varying by crop.
- Isolating Wheat alone reveals genuine signal: Fertilizer correlates at +0.447 and Rainfall at -0.410 — wheat is a winter crop that benefits from fertilizer input but performs worse under high rainfall, both agronomically expected relationships invisible in the mixed-crop analysis.

KPI Summary

KPI Metric	Value	KPI Metric	Value
Usable Records (Excl. Coconut)	19,480	Year Range	1997–2019

KPI Metric	Value	KPI Metric	Value
States / Crops Covered	30 / 54	Total Production	14,845,100,222
Avg Yield	4.41	Coconut % of Raw Production	95.41%
Top State	Uttar Pradesh — 4.44bn	2nd State	Maharashtra — 1.88bn
Top Crop by Area	Rice — 992M hectares	2nd Crop	Wheat — 602M hectares
Top Yield State	Delhi — 13.12	Whole Year Crop Yield	13.23 (vs Kharif 2.48)
Fertilizer Growth (1997-2019)	+53.2%	Yield Growth (Same Period)	+9.9%
Wheat — Fertilizer Correlation	+0.447	Wheat — Rainfall Correlation	-0.410
District-Level Records	245,529 (646 districts)	Top District	Muzaffarnagar, UP — 269M

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Choosing Between Two Non-Overlapping Datasets

- Problem: Unlike a prior project where two files were different formats of identical data, these two files had genuinely different granularities (state vs district), different crop coverage (55 vs 124), and no clean shared key for a direct join — a naive single-table approach would have discarded one file's unique value entirely.
- Solution: Built a 2-table MySQL model preserving each file's distinct strength — the state-level file for input-driver analysis, the district-level file for geographic depth — the first JOIN-capable architecture in this portfolio, reflecting the data's genuine structure rather than forcing an artificial merge.

Challenge 2 — Silent NULL-to-Zero Corruption via Hidden Carriage Return

- Problem: After fixing an initial strict-mode import error, a verification query expecting 3,730 null Production values returned 0. Root cause: the CSV's Windows \r\n line endings combined with LINES TERMINATED BY '\n' left a trailing \r on the last-loaded field for every row. For blank rows, this meant the field held the single character \r rather than true emptiness — and MySQL's non-strict CAST() of an unparseable string to DECIMAL silently returns 0 rather than erroring, with genuinely-reported zero-production rows sharing that same final value and making the corruption invisible at a glance.
- Solution: Diagnosed by tracing one specific known-blank row byte-for-byte back to the raw CSV file, confirming the exact \r mechanism before attempting a fix. Rebuilt the import using a VARCHAR staging column and an explicit TRIM(TRAILING '\r' FROM ...) conversion in a standalone UPDATE statement — verified against the exact same known-blank row afterward to confirm it now correctly returned NULL, and confirmed the total null count matched Python's original count of 3,729 exactly.

Challenge 3 — Reserved-Word Column Naming

- Problem: The source column name "Yield" risked ambiguity with MySQL's handling of certain reserved-adjacent identifiers in some contexts.

- Solution: Proactively named the column `yield_value` in the schema design rather than encountering the issue mid-import, avoiding any need for backtick-escaping throughout every subsequent query.

11 SKILLS LEARNED & TAKEAWAYS

- Recognizing Unit-of-Measurement Distortions: Learned to treat an implausibly dominant single-category result (95%+ of a total from under 1% of rows) as a diagnostic signal demanding investigation, not an interesting finding to report at face value.
- Multi-Table Architecture Decisions: Practiced recognizing when two datasets are genuinely complementary rather than duplicate formats, and building a 2-table model that preserves each source's unique analytical value instead of forcing a lossy merge.
- Tracing Corruption to Root Cause: Diagnosed a second occurrence of the hidden-carriage-return bug pattern in a completely different context (numeric CAST silently defaulting to zero, rather than text categorization defaulting to a catch-all) — reinforcing that the same underlying mechanism can manifest differently depending on the target data type.
- Distinguishing State-Year Aggregates from Row-Level Data: Recognized that Fertilizer, Pesticide, and Rainfall values repeat identically within a state-year across all crops, and that meaningful correlation analysis requires isolating a single crop rather than mixing dissimilar units together.
- Verifying Fixes Against Independent Sources: Cross-checked every MySQL query result against Python's independently generated charts throughout — catching that a "successful" import can still silently corrupt data even when it produces no errors and returns a plausible-looking row count.

12 CONCLUSION

- Processed two complementary Indian agricultural datasets through a structured Python and dual-table MySQL pipeline — removing incomplete trailing years, engineering boundary-safe derived features, and building a 2-table, JOIN-capable database model that preserves each source's distinct analytical strength.
- Diagnosed and corrected two independent, non-obvious data quality issues: a 95%+ unit-of-measurement distortion from Coconut's nut-count convention, and a silent NULL-to-zero corruption from a hidden Windows carriage-return character — both verified via byte-level tracing back to the original source files.
- Surfaced that fertilizer use grew 53.2% while yield grew only 9.9% (a genuine diminishing-returns finding), that Uttar Pradesh leads national production at 4.44 billion, and that isolating a single crop (Wheat) reveals real input-yield correlations invisible in mixed-crop analysis.
- Delivered a 3-page, 21-visual Power BI dashboard including a true geographic map, alongside a portfolio-grade custom JSON theme (India Agricultural Intelligence — Harvest Gold), demonstrating end-to-end command of dual-source data integration, MySQL debugging, and DAX-based aggregation correction.